

End of topic test

Revise 360 lesson plan — 60 minutes

Unit	OCR J277 2.1 Algorithms — Paper 2, Computational thinking, algorithms and programming
Lesson	End of topic test (lesson 12 of the unit)
Lesson ID	al-l12-test
Experience ID	none — this lesson is on paper
Plan ID	PLAN-AL-L12-TEST
Content version	7caa3ea54fbe
Website	no online route — this lesson is on paper

Big question, outcomes and success criteria

this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.

Prerequisites

The lesson before this one is al-l11. Pupils should already have finished it, and everything earlier in the unit: this lesson assesses all of it.

Earlier in the unit: al-l01, al-l02, al-l03, al-l04, al-l05, al-l06, al-l07, al-l08, al-l09, al-l10.

Assessable outcomes

this lesson states no outcomes of its own, which is correct for what it is: a test and reflection sheet. It assesses the outcomes taught across the unit.

Vocabulary, resources and preparation

No new vocabulary: this lesson introduces nothing new. The vocabulary it draws on is listed on the earlier lessons' plans and in the unit Big Picture.

Resources

WORKSHEET worksheets/AL_L12_EndOfTopicTest_Reflection_Worksheet.docx

Before the lesson

1. Print the paper, one per pupil, and the reflection sheet to hand out afterwards.
2. Have the answer sheet to hand. It is teacher-only and is in the unit teacher guide pack, not in this plan.
3. Decide whether you are marking in the lesson or collecting the papers in.

Nothing online is needed. Pupils do not sign in for this lesson, and there is no experience to launch.

Teaching sequence

Sixty minutes including transitions. The test itself is fixed; if your lesson is shorter, mark it afterwards rather than cutting the test.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
0-5	Settle and set up	Hand out the paper. Remind the class that this is from memory and in silence.	Put bags away and take out a pen.	Everyone has a paper and nothing else.

MIN	PHASE	TEACHER	PUPILS	EVIDENCE TO CHECK
5-45	The test	Silent, timed. Collect every paper.	Answer from memory.	Silence, and every paper collected.
45-55	Mark or self-mark	Against the answer sheet, if you are marking in the lesson; otherwise collect and mark after.	Mark against the answer sheet.	A mark per question, not just a total.
55-60	Set the reflection	Issue the reflection worksheet and explain that it is completed once the test is returned.	Write their weakest topic on the reflection sheet.	The reflection sheet names a topic, not a feeling.

The explanation and the worked example

This lesson teaches nothing new and has no walls to read: it is the unit test and its reflection sheet. The explanations it rests on are in the plans for the lessons before it.

Retrieval and the check for understanding

This lesson is the unit test and its reflection sheet. It has no starter and no hinge question by design: the whole hour is the check. The retrieval it rests on is everything taught in the unit, and the reflection sheet is where a pupil records which topics to go back to.

What the test covers

1. Abstraction
2. Decomposition and structure diagrams
3. Algorithmic thinking, inputs, processes and outputs
4. Flowcharts and their symbols
5. Pseudocode and the OCR reference language
6. Linear search
7. Binary search
8. Bubble sort
9. Merge sort and insertion sort
10. Syntax and logic errors
11. Trace tables

Mark against the unit answer sheet, which is teacher-only and not printed here. Give a mark per question rather than a total: the reflection sheet needs the breakdown to be worth filling in.

Guided to independent practice

There is no guided practice in this lesson, by design: every question is answered independently, from memory and in silence. The guided and independent practice it assesses happened in the lessons before it.

Accessibility

- Read the paper aloud for any pupil whose access arrangements allow it.
- Extra time applies here as it would in any assessment.
- A pupil who cannot write at length may answer the shorter questions first; the marks are per question, so nothing is lost by the order.

Delivering the online experience

This lesson has no online experience: it is a test and its reflection sheet. Everything it needs is on paper.

Assessment, evidence and unfinished work

This lesson is the assessment. Every question is marked against the unit answer sheet, which is teacher-only and in the unit teacher guide pack, not on this plan.

ID	QUESTION	MARKS	WHERE
al-l12-test-x1	Abstraction	—	the unit test, question 1; the reflection worksheet, My results
al-l12-test-x2	Decomposition and structure diagrams	—	the unit test, question 2; the reflection worksheet, My results
al-l12-test-x3	Algorithmic thinking, inputs, processes and outputs	—	the unit test, question 3; the reflection worksheet, My results
al-l12-test-x4	Flowcharts and their symbols	—	the unit test, question 4; the reflection worksheet, My results
al-l12-test-x5	Pseudocode and the OCR reference language	—	the unit test, question 5; the reflection worksheet, My results
al-l12-test-x6	Linear search	—	the unit test, question 6; the reflection worksheet, My results
al-l12-test-x7	Binary search	—	the unit test, question 7; the reflection worksheet, My results
al-l12-test-x8	Bubble sort	—	the unit test, question 8; the reflection worksheet, My results
al-l12-test-x9	Merge sort and insertion sort	—	the unit test, question 9; the reflection worksheet, My results
al-l12-test-x10	Syntax and logic errors	—	the unit test, question 10; the reflection worksheet, My results
al-l12-test-x11	Trace tables	—	the unit test, question 11; the reflection worksheet, My results

Expected answers

Model answers and marking guidance are teacher-only and are not printed on this plan. They are in the unit's teacher guide pack, and the per-activity marking for this lesson is in `answers/records/al-l12-test.json`, which is not published with the website.

Common errors

None are recorded for this lesson beyond the misconceptions listed above.

An unfinished paper

Mark what is there and record the questions left blank as blank, not as wrong: a pupil who ran out of time and a pupil who could not answer need different responses, and the reflection sheet cannot tell them apart if the marking does not.

Exit, next lesson and homework

Exit check

The test is the check. The reflection sheet is the follow-up: a pupil writes their mark for each question, colours the red/amber/green column and names what they will do about each topic they lost marks on.

Retrieval link to next lesson

Next is `al-bonus`, Algorithm arena. Use this lesson's big question as next lesson's starter, so the first thing they do is retrieve it.

After this lesson

Return the marked papers and complete the reflection sheet with the class. Set the two or three experiences a pupil's weakest topics point to: the reflection sheet names them, and review mode on those lessons reopens the questions they got wrong.

Shorter lessons, longer lessons and no internet

A shorter lesson

Do not cut the test. Run it to its full length and mark it afterwards; set the reflection sheet for the start of the next lesson.

If the internet or the devices are unavailable

It makes no difference to this lesson. Everything it needs is on paper.

The reflection sheet points pupils back at the experiences for the topics they lost marks on; set those when the devices are back.

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